

IA 640 - Project 2

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The dataset we selected is 'Walmart Customer Purchases' sourced from Kaggle. It includes transactions for the year 2024-2025. The aim of this project is to visualize Walmart sales of different categories of products based on customer demographics and transaction details.

The dataset consists of variables such as:

Categorical variables - Customer_Id, Gender, City, Category, Product_Name, Purchase_Date, Payment_Method, Discount_Applied, Repeat_Customer

Quantitative variables - Age, Purchase_Amount, Rating

To prepare the data for visualization, the Customer_Id column was removed since it holds unique values and does not contribute meaningful insights for group analysis.

Additionally, a new categorical variable named AgeGroup was created by binning the continuous Age variable into different ranges. The user interface allows users to apply filters based on Age Group, City, Gender, Repeat Customer status, and whether a Discount was applied. Each filter can be enabled or disabled using corresponding checkboxes. This design provides flexibility to either analyze the entire dataset or focus on specific segments of interest. The main output of the application is a bar plot, which dynamically updates to show the total purchase amount by product category, depending on the filters applied.

This project successfully demonstrates the use of Shiny and Shinyjs to create a responsive and interactive dashboard. One of the highlights was the ability to toggle filters on and off in real-time, showcasing the power of reactive programming in R. A key challenge encountered was managing the interplay between multiple filters especially ensuring that user selections were respected when some filters were disabled, and that the reactive data pipeline updated correctly without breaking. Designing a clean and consistent user experience while maintaining data accuracy and responsiveness required thoughtful logic implementation and careful testing. Overall, the project enhanced understanding of Shiny's reactive environment, UI customization, and data manipulation techniques.

ShinyApp_Link

Walmart 2024-25 Sales by Category

☐ Disable

Select Age Group

26-3536-45

☒ Disable

Select City

Johnhaven

☐ Disable

Select Gender

Male

☐ Disable

Repeated Customer

No

☐ Disable

Discount Applied

Yes

